

SIMATS ENGINEERING

ASSESSMENT TOOL 1-Pseudocode Writing Task

COURSE NAME: R PROGRAMMING

COURSE CODE: ITA04

COURSE OUTCOME COVERED

CO: Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively.
(Bloom Level -BL2- BL4)

Assessment Tool Weightage: 20%

QUESTIONS:

Total Marks: 100

1. Variables and Data Types Analysis (BL4 – Analyze)

Design pseudocode to simulate basic variable handling in R programming:

- Declare variables of different data types (numeric, character, logical)
- Perform arithmetic and logical operations
- Display the results

START

```
Declare Num = 50.5      // Numeric
Declare Name = "Godwin" // Character
Declare Status = TRUE   // Logical
```

```
Display Num
Display Name
Display Status
```

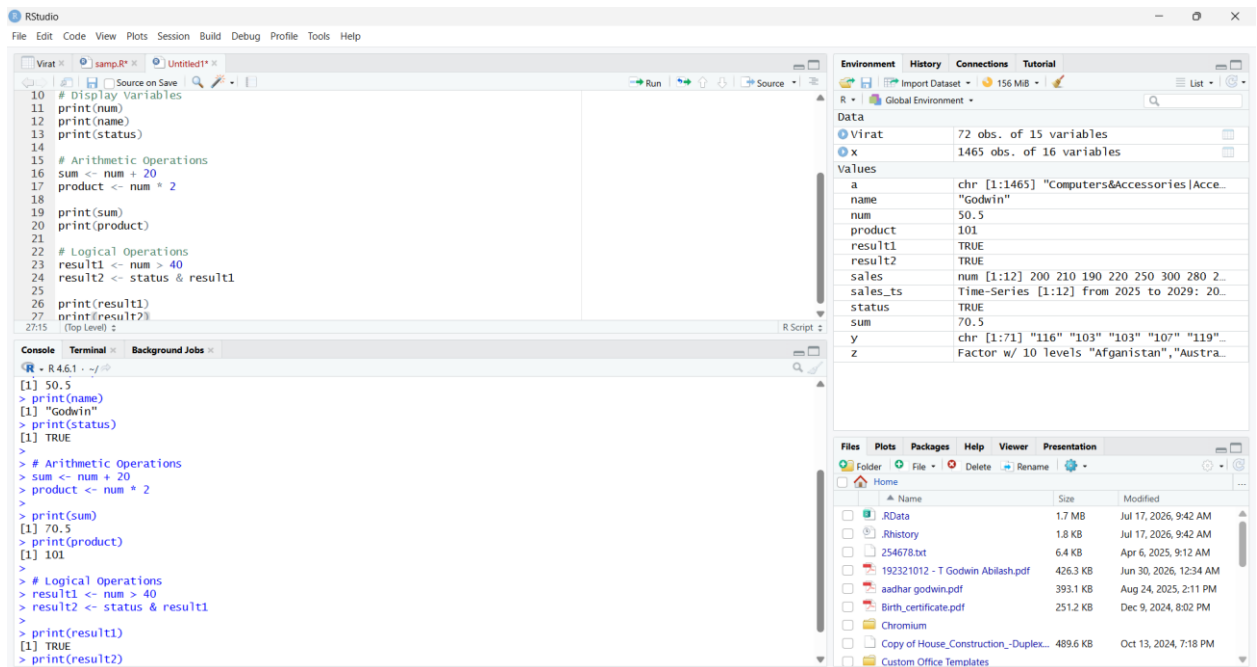
```
// Arithmetic Operations
Sum = Num + 20
Product = Num * 2
```

```
Display Sum
Display Product
```

```
// Logical Operations
Result1 = (Num > 40)
Result2 = Status AND Result1
```

```
Display Result1
Display Result2
```

STOP



2. Data Structures Implementation (BL4 – Analyze)

Write pseudocode to demonstrate advanced data structures in R:

- Create and initialize a vector, list, and matrix
- Perform operations such as indexing, updating, and aggregation
- Compare the structures based on usage

START

```
// Create Vector
Vector = [10,20,30,40,50]
```

```
// Create List
List = ["Godwin",21,TRUE]
```

```
// Create Matrix
Matrix =
10 20
30 40
```

// Indexing

Display Vector[3]

Display List[1]

Display Matrix[2][1]

// Updating

Vector[2] = 25

Matrix[1][2] = 50

// Aggregation

Sum = Sum of Vector Elements

Display Sum

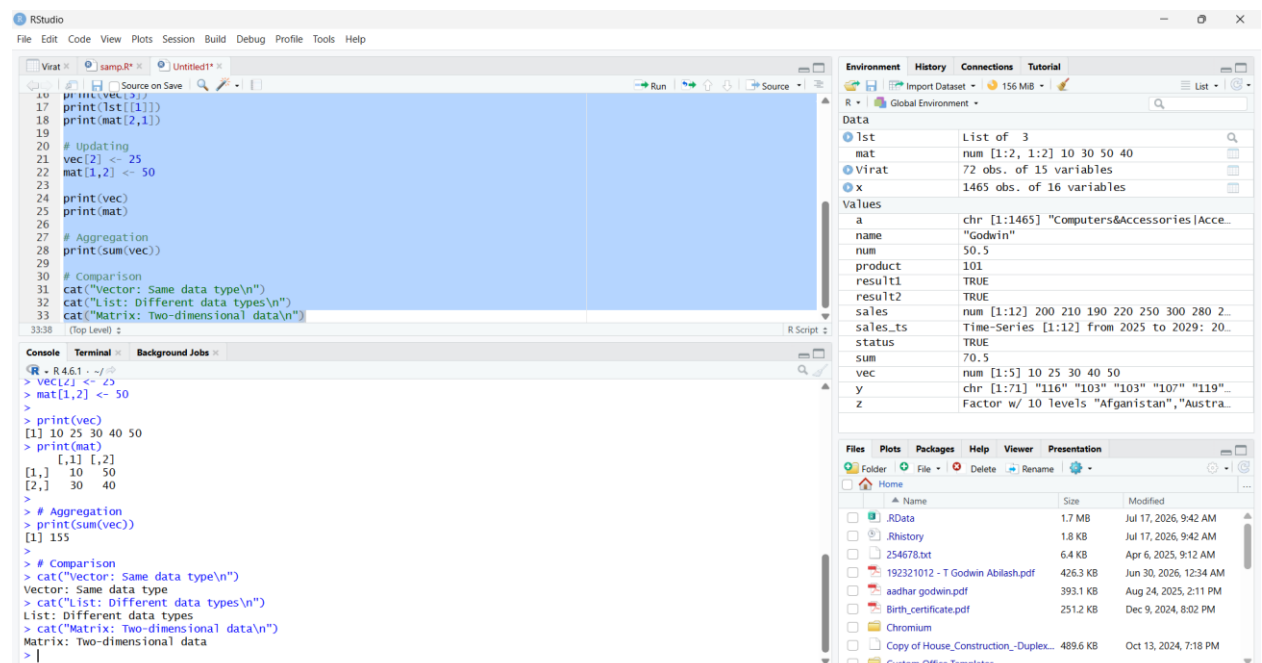
// Comparison

Display "Vector stores same type data."

Display "List stores different data types."

Display "Matrix stores data in rows and columns."

STOP



3. Matrix and Array Operations (BL4 – Analyze)

Design pseudocode to perform operations on matrices and arrays:

- Create a matrix and a 3D array
- Perform addition, multiplication (matrix), and element-wise operations
- Display the results and analyze differences

START

Create MatrixA =

1 2

3 4

Create MatrixB =

5 6

7 8

Create 3D Array

Array[2][2][2]

// Matrix Addition

MatrixC = MatrixA + MatrixB

// Matrix Multiplication

MatrixD = MatrixA $\times$ MatrixB

// Element-wise Multiplication

MatrixE = MatrixA * MatrixB

Display MatrixC

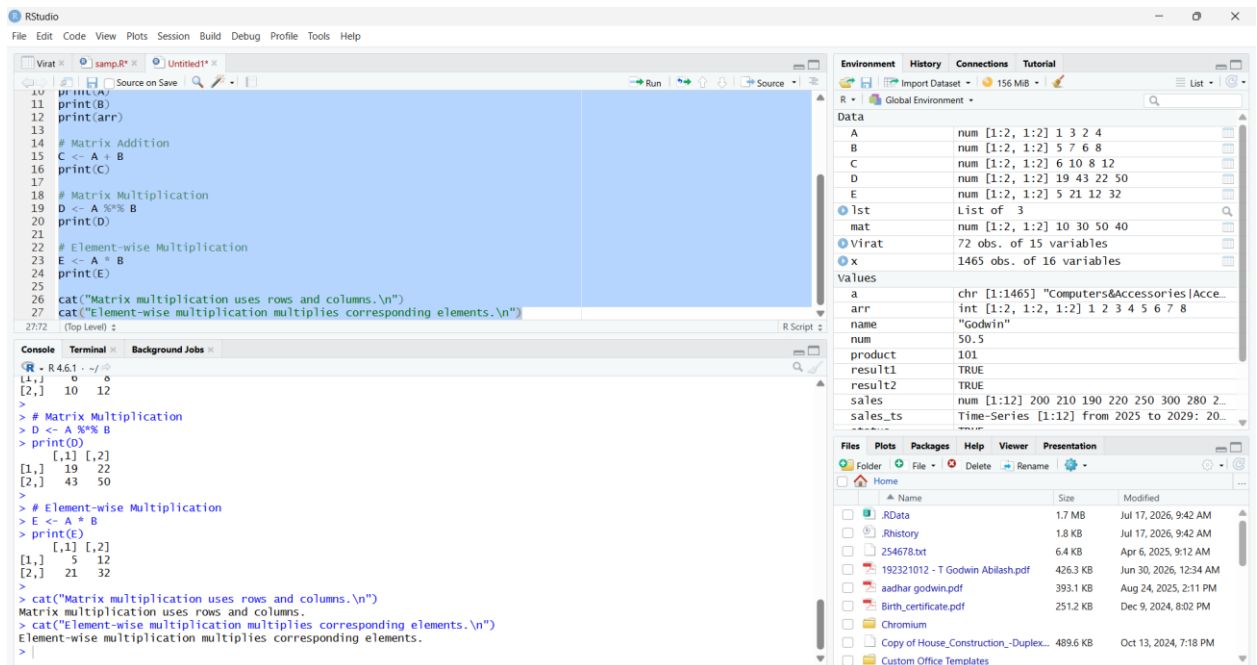
Display MatrixD

Display MatrixE

Display "Matrix multiplication follows row-column rule."

Display "Element-wise multiplication multiplies corresponding elements."

STOP



4. Data Frame Construction and Manipulation (BL4 – Analyze))

A dataset needs to be organized efficiently.

Write pseudocode to to:

- Create a data frame with multiple columns (numeric and character)
- Add, delete, and modify columns
- Perform basic summary operations

START

Create DataFrame

Column Name = ["A","B","C"]

Column Age = [20,21,22]

Column Department = ["IT","CSE","ECE"]

Display DataFrame

// Add Column

Add Marks = [85,90,95]

// Modify Column

Age[2] = 23

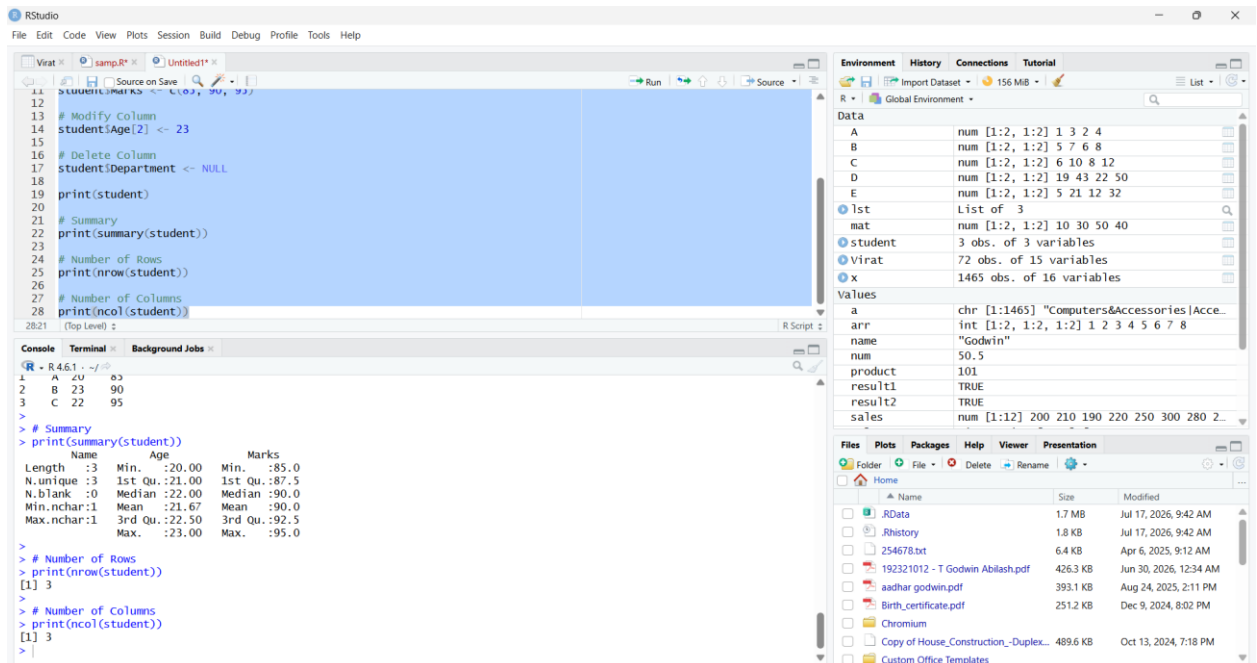
// Delete Column

Delete Department

Display Updated DataFrame

Display Number of Rows
Display Number of Columns
Display Summary of Numeric Columns

STOP



5. Functions and Logical Operations (BL4 – Analyze)

Design pseudocode to implement functions and logical vector operations:

- Create a user-defined function with input parameters
- Perform operations on logical vectors
- Return and display computed results

START

Function Square(Number)

Result = Number × Number

Return Result

End Function

Value = Square(6)

Display Value

LogicalVector = [TRUE,FALSE,TRUE,TRUE,FALSE]

TrueCount = Count TRUE values

Display TrueCount

If TrueCount > 2 Then

Display "Majority are TRUE"

Else

Display "Majority are FALSE"

End If

STOP

The screenshot shows the RStudio interface. The script editor on the left contains the following R code:

```
11 # LOGICAL VECTOR
12 logic <- c(TRUE, FALSE, TRUE, TRUE, FALSE)
13
14 print(logic)
15
16 # Count TRUE values
17 count <- sum(logic)
18
19 print(count)
20
21 if(count > 2)
22 {
23   print("Majority are TRUE")
24 }
25 else
26 {
27   print("Majority are FALSE")
28 }
```

The console on the bottom left shows the execution output:

```
> print(result)
[1] 36
>
> # Logical Vector
> logic <- c(TRUE, FALSE, TRUE, TRUE, FALSE)
>
> print(logic)
[1] TRUE FALSE TRUE TRUE FALSE
>
> # Count TRUE values
> count <- sum(logic)
>
> print(count)
[1] 3
>
> if(count > 2)
+ {
+   print("Majority are TRUE")
+ }
+ [1] "Majority are TRUE"
> else
```

The Environment pane on the right shows the variables created:

Variable	Class	Length	Values
a	chr	[1:1465]	"Computers&Accessories Acce...
arr	int	[1:2, 1:2, 1:2]	1 2 3 4 5 6 7 8
count	3L		
logic	logi	[1:5]	TRUE FALSE TRUE TRUE FALSE
name	"Godwin"		
num	50.5		
product	101		
result	36		
result1	TRUE		
result2	TRUE		
sales	num	[1:12]	200 210 190 220 250 300 280 2...
sales_ts	Time-Series	[1:12]	from 2025 to 2029: 20...
status	TRUE		
sum	70.5		
vec	num	[1:5]	10 25 30 40 50
y	chr	[1:71]	"116" "103" "103" "107" "119"
z	Factor w/ 10 levels		"Afghanistan", "Austra...

The Files pane on the bottom right shows the file explorer with various files and folders.

Code of Conduct:

I T Godwin Abilash (192321012) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient